Exercise Questions

Question 1

Suppose we have a simplified language with:

- Three vowels: a, i, and u
- Three consonants: p, t, and k

Let X map consonants $\{p, t, k\}$ to range $R_X = \{1, 2, 3\}$
Let Y map vowels $\{a, i, u\}$ to range $R_Y = \{1, 2, 3\}$

Suppose we estimated the joint PMF for vowel and consonant co-occurrences within syllables, as given below.

$$P(x, y) = \begin{array}{c|c|c|c} x \setminus y & 1 & 2 & 3 \\ \hline 1 & \frac{1}{16} & \frac{3}{8} & \frac{1}{16} \\ 2 & \frac{1}{16} & \frac{3}{16} & 0 \\ 3 & 0 & \frac{3}{16} & \frac{1}{16} \end{array}$$

Questions: Calculate the following:

- Entropies $H(X)$ and $H(Y)$
- Conditional entropies $H(X|Y)$ and $H(Y|X)$
- Joint entropy $H(X, Y)$
- Mutual information $I(X; Y)$

Question 2: From the 2021 AI2 exam paper

As a data scientist in a telecommunication company, your task is to analyze a customer dataset to predict whether a customer will terminate his/her contract. The dataset consists of around 8000 customer records, each consisting of one binary dependent variable Y, indicating whether the customer terminates the contract ($Y = 1$) or not ($Y = 0$), and 19 independent variables, which include the customer's information, e.g., age, subscription plan, extra data plan, etc., and consumer behavior such as the average number of calls and hours per week.

Since your boss needs some actionable insights to retain customers, you decided to use interpretable machine learning methods. Design your interpretable machine learning method by answering the following questions:

(a)

You have implemented a feature selection algorithm based on mutual information to select the most informative features from the 19 independent variables. To validate the implementation of your mutual information calculation function, you use a small subset of the data to calculate mutual information manually. You select one independent variable *subscription plan*, denoted as S, which takes two values, $S \in \{1, 2\}$. Please use the following Probability Mass Function table:

$p(S, Y)$	$S = 1$	$S = 2$
$Y = 0$	$\frac{2}{12}$	$\frac{5}{12}$
$Y = 1$	$\frac{2}{12}$	$\frac{3}{12}$

Calculate:

- Entropies $H(S)$ and $H(Y)$
- Conditional entropies $H(S|Y)$ and $H(Y|S)$
- Joint entropy $H(S, Y)$
- Mutual information $I(S; Y)$

Show all your working. Discuss what mutual information means and whether this feature will be selected or not.

(b)

After applying your algorithm, you selected two variables:

- Extra data plan E, which is a binary random variable that indicates whether the customer subscribes to the extra data plan ($E = 1$) or not ($E = 0$).
- Averaged hours used per week H, which is a continuous random variable.

You then built a logistic regression model to classify customers into *low risk* or *high risk* of terminating the contract. The fitted model is:

$$\log\left(\frac{p}{1-p}\right) = -0.77 + 0.23H - 1.18E$$

i) Given a customer $\mathbf{x}$ who has the extra data plan ($E = 1$) and spent on average 0.5 hours per week, calculate the odds and the probability that the customer will terminate the contract ($Y = 1|\mathbf{x}$).
ii) Using this fitted model, explain to your boss what actions should be taken to retain customers.

Question 3: From the 2022 AI2 exam paper

As a machine learning expert for an AI cybersecurity company, your task is to design an automated network intrusion detection system. You have collected a large number of records of network activities. Each record includes the log information about network activity, such as protocol types, duration, number of failed logins, which are random variables, denoted as $\mathbf{X} = [X_1, X_2, \ldots, X_n]$. Each record also includes a binary random variable Y called label that was labeled by cybersecurity experts as intrusions ($Y = 1$) or normal connections ($Y = 0$).

Answer the following questions about Feature Selection Based on Mutual Information.

(i) Explain to your colleague, who knows nothing about information theory, the concept of mutual information.

(ii) Explain the loop in the pseudocode of Table 1.

1.	Initialize: Set $F \leftarrow \mathbf{X}$ and $S \leftarrow \emptyset$
2.	Select $f_{max} = \arg\max_{X_i \in X} I(Y; X_i)$
3.	Update $F \leftarrow F \setminus \{f_{max}\}$, $S \leftarrow \{f_{max}\}$
4.	Repeat until $\lvert S \rvert = K$:
5.	Select $f_{max} = \arg\max_{X_i \in F} \left[I(Y; X_i) - \sum_{X_s \in S} I(X_s; X_i) \right]$
6.	Update $F \leftarrow F \setminus \{f_{max}\}$, $S \leftarrow S \cup \{f_{max}\}$

Table 1: Pseudocode of Mutual Information based Feature Selection Algorithm

Question 4: Decision Tree Classification

A company wants to classify whether a customer will purchase a product (Y = Yes) or not (Y = No) based on categorical features. You have access to the following dataset:

Age Group	Income Level	Prior Purchase	Purchase (Y)
Young	Low	No	No
Young	Low	Yes	No
Young	High	No	Yes
Middle-aged	Low	No	No
Middle-aged	High	No	Yes
Senior	Low	No	No
Senior	High	Yes	Yes
Senior	High	No	Yes

(a) Calculate the entropy of the target variable Y.

(b) Compute the information gain for splitting on the feature "Prior Purchase."

(c) If you were to build a decision tree, which feature would be the best root node? Justify your answer.

(d) Draw the first split of the decision tree based on the best attribute.

(e) Discuss how overfitting can be avoided in decision trees and suggest techniques to improve generalization.